Data Dictionary

MIMIC-III Clinical Database (v1.4)
Bayesian Hierarchical Survival Modeling

PHP2530 Final Project

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1 Overview and Data Structure

This data dictionary documents all variables extracted from the MIMIC-III Clinical Database for Bayesian hierarchical survival modeling. The preprocessing script (`data-prep.R`) creates intermediate datasets and **final analytical objects** for modeling.

Primary Outputs (loaded via `source("load.R")`):

- `X`: Static baseline predictor matrix ($N \times 66$)
- `y`: Binary ICU mortality outcome vector (length N)
- `S_mat`: Time-varying predictor matrix ($\sum T_i \times Q$)
- `y_S`: Discrete-time outcome vector (length $\sum T_i$)

See Section 3 for comprehensive documentation of these objects.

1.1 Conceptual Framework

For patient $i = 1, \ldots, N$ with T_i discrete time periods, we define three **intermediate data frames** (for exploration) and two **final analytical objects** (for modeling):

Intermediate Data Frames (retain all variables, including IDs and outcomes):

1. **Static Baseline Covariate Matrix X** `(ready_df)` $\longrightarrow$ cleaned to `X, y`
   - Dimension: $N \times P$ where N = number of admissions, P = number of covariates
   - Each row represents ONE admission at "Time Zero" (ICU admission)
   - Contains: demographics, aggregated vitals/labs, comorbidities, procedures, interventions
   - These covariates are **constant within patient** — they do not change over time
2. **Time-Varying Covariate Matrix $\mathbf{Z}_{i,t}$** `(longitudinal_df)`
   - Structure: One row per measurement occasion (patient $\times$ time point)
   - $Z_{i,t}$ = covariate vector for patient i at time t
   - Contains: vital signs (HR, BP, SpO2, temp, RR) with precise timestamps
   - These covariates **change over time** — different values at each t
3. **Discrete-Time Survival Format** `(person_period_df)` $\longrightarrow$ cleaned to `S_mat, y_S`
   - Dimension: $\sum_{i=1}^{N} T_i$ rows (one row per patient per period at risk)
   - Structure: (i, t) indexed — patient i, period t
   - Contains: discrete outcome $y_{i,t}$, static covariates X_i, period-aggregated $Z_{i,t}$
   - Designed for Polya-Gamma augmented Gibbs sampling

1.2 Data Source Options

The preprocessing script supports two data sources via `DATA_SOURCE`:

- `DATA_SOURCE <- "demo"`: MIMIC-III Demo (~100 patients) — for development
- `DATA_SOURCE <- "full"`: Complete MIMIC-III from PhysioNet (~46,000 patients)

1.3 Output Files

All files saved to `data/rds/` as R serialized objects:

File	Description
Intermediate Data Frames (for exploration/debugging)	

File	Description
`ready_df.rds`	Static baseline matrix before final exclusions
`longitudinal_df.rds`	Time-varying measurements $\mathbf{Z}_{i,t}$
`person_period_df.rds`	Discrete-time survival format (raw)
Final Analytical Objects (primary use — loaded by `load.R`)	
`X_y.rds`	List: {`X, y`} — Static predictor matrix and outcome
`S_y.rds`	List: {`S, y`} — Time-varying matrix and discrete outcome

Important: The final analytical objects (`X_y.rds` and `S_y.rds`) are the primary outputs for modeling. The intermediate data frames are retained for exploration but have NOT undergone final cleaning (exclusion of IDs, outcomes, multicollinear variables, etc.).

2 Time Zero and Survival Framework

2.1 Time Zero Definition

Time Zero is defined as the **first ICU admission** within each hospital stay (`icu_intime`). This establishes a coherent survival framework:

Component	Definition
Time origin	First ICU admission timestamp (`icu_intime`)
Event	Death during hospitalization (`event_status = 1`)
Censoring	Discharge alive (`event_status = 0`)
Time scale	Hours from ICU admission to death/discharge

Justification: The ICU admission marks entry to highest-acuity care. Survival from this point is the clinically relevant outcome. Using the *first* ICU stay avoids survivorship bias from conditioning on patients who survived long enough for subsequent transfers.

2.2 Continuous-Time Outcome

For the continuous-time framework (`ready_df`):
- `time_to_event_hours`: T_i = hours from ICU admission to event/censoring
- `event_status`: $\delta_i = 1$ if died, $\delta_i = 0$ if censored

2.3 Discrete-Time Outcome

For the discrete-time framework (`person_period_df`), continuous time is discretized into 24-hour periods:
- Period 1: hours 0–24, Period 2: hours 24–48, etc.
- Maximum 7 periods (capped for computational tractability)
- `y_it`: binary outcome — 1 if event in period t, 0 otherwise

Example:

Patient Type	$y_{i,1}$	$y_{i,2}$	$y_{i,3}$	$y_{i,4}$	Interpretation
Dies in period 3	0	0	1	—	Event in final period
Censored in period 4	0	0	0	0	No event (discharged alive)

3 Final Analytical Objects (Primary Outputs)

This section documents the **primary analytical objects** produced by `data-prep.R` and loaded by `load.R`. These are the clean, analysis-ready matrices and vectors for Bayesian modeling.

3.1 Loading the Data

Use `source("load.R")` to load all final objects into your R environment:

```
source("load.R")
# Objects now available:
#   X      - Static baseline predictor matrix (N x P)
#   y      - Binary outcome vector (length N)
#   S_mat  - Time-varying predictor matrix (sum of T_i x Q)
#   y_S    - Discrete-time outcome vector (length sum of T_i)
```

3.2 Object Relationship Diagram

Static Analysis		**Discrete-Time Analysis**
$\mathbf{X}$ (123×66)	$\longrightarrow$	$\mathbf{S}$ $(\sum T_i \times \mathrm{Q})$
$\downarrow$		$\downarrow$
$\mathbf{y}$ (123 elements)		$\mathbf{y}_S$ $(\sum T_i$ elements)
One row per admission		*One row per person-period*
BART, logistic regression		*Polya-Gamma Gibbs sampling*

3.3 X: Static Baseline Predictor Matrix

Dimensions: $N \times P$ where N = number of admissions (after complete-case filtering), P = number of predictors (66 columns in demo data).

Source: Derived from `ready_df` after:

1. Exclusion of IDs, outcomes, timestamps, and redundant variables (Section 8.4)
2. Ethnicity recoding to numeric factors (Section 7.10)
3. Complete-case analysis (`na.omit()`)
4. Removal of `hadm_id` and `icu_mortality` (extracted as `y`)

Column Categories in X:

Category	**Variables Included**
Categorical Factors (Multi-level)	
Demographics	admission_type, insurance, ethnicity (0–7), gender, first_careunit
Clinical Factors	age_group (Adult/Elderly/Very Old), los_group (Short/Prolonged/Extended), gcs_status, complexity_status, comorbidity_level
Vital Sign Status	hr_status, rr_status, spo2_status, temp_status
Lab Status	bun_status, creat_status, glucose_status, hgb_status, plt_status, k_status, na_status, wbc_status
Binary Indicators (as Factors: 0/1)	

Category	Variables Included
Comorbidities	icd_sepsis, icd_heart_failure, icd_aki, icd_ckd, icd_resp_failure, icd_pneumonia, icd_diabetes, icd_liver_disease, icd_malignancy, icd_stroke, icd_mi, icd_copd
Interventions	is_ventilated, received_dialysis, has_central_line, received_transfusion, is_surgical, on_antibiotic, on_anticoagulant, on_insulin, on_sedative
Microbiology	positive_blood_culture, any_positive_culture
Other	multiple_icu_stays, callout_requested, hr_unstable, temp_unstable
Continuous (Retained)	
Stay Characteristics	los_icu, n_icu_stays
Counts	n_diagnoses, n_procedures, n_prescriptions
Acuity	gcs_mean_val, vasopressor_duration_hrs, drg_severity, drg_mortality

3.4 y: Outcome Vector

Dimensions: Length N (same as number of rows in `X`).

Definition: `icu_mortality` — binary indicator of ICU mortality.

- `y = 1`: Patient died during ICU stay
- `y = 0`: Patient survived ICU stay (discharged alive or transferred)

Relationship to other outcomes:

- More conservative than `hospital_expire_flag` (hospital-wide mortality)
- Temporally bounded to the ICU admission window
- Used as the target for BART classification and static logistic regression

3.5 S_mat: Time-Varying Predictor Matrix

Dimensions: $\left(\sum_{i=1}^{N} T_i\right) \times Q$ where T_i = number of discrete periods patient i is at risk, Q = number of time-varying predictors.

Source: Derived from `person_period_df` after:

1. Filtering to matched cohort (`hadm_id %in% valid_hadm_ids`)
2. Same variable exclusions as `X`
3. Ethnicity recoding to match `X`
4. Complete-case analysis (`na.omit()`)
5. Removal of `y_it`, `icu_mortality`, and `hadm_id`

Key Variable: `period` (integer 1–7) is **retained** in `S_mat` to support baseline hazard approximations in discrete-time models.

Structure: Each patient contributes T_i rows (one per period at risk). Static covariates are repeated across periods; time-varying vitals are period-specific aggregates (mean, min, max per 24-hour window).

Example rows for one patient:

row	period	age_group	hr_mean	temp_mean	...
1	1	Elderly	85.2	37.1	...
2	2	Elderly	88.7	37.3	...
3	3	Elderly	92.1	37.8	...

3.6 y_S: Discrete-Time Outcome Vector

Dimensions: Length $\sum_{i=1}^{N} T_i$ (same as number of rows in `S_mat`).

Definition: `y_it` — binary indicator of event in period t for patient i.

- `y_S[row] = 1`: Event (death) occurred in this period
- `y_S[row] = 0`: No event in this period (survived to next period or censored)

Key property: For each patient, `y_S = 1` in **at most one row** (the final period if they died). All prior periods have `y_S = 0`.

Use case: Polya-Gamma augmented Gibbs sampling for discrete-time survival models:

$$\text{logit}\, P(y_{it} = 1 \mid y_{i,t-1} = 0) = \alpha_t + S'_{it}\beta$$

3.7 Correspondence Between Objects

Static Object	Time-Varying Object	Relationship
`X`	`S_mat`	Static vars repeated; S adds `period` + time-varying vitals
`y`	`y_S`	`y` = patient-level; `y_S` = period-level (sums to same # even
`hadm_id` (row names)	`hadm_id` (in S before drop)	Links patients across matrices

4 Static Baseline Matrix X (ready_df)

This dataset contains **one row per hospital admission**. All variables represent information available at or aggregated across the ICU stay, treated as "baseline" for modeling purposes.

4.1 Identification Variables

Variable	Type	Description
subject_id	Factor	Unique patient identifier (for patient-level random effects)
hadm_id	Factor	Unique hospital admission identifier (unit of analysis)
icustay_id	Integer	First ICU stay identifier within admission

Note: All identification variables are **excluded** from the final predictor matrix `X` in Section 17. They are retained in `ready_df` for reference and hierarchical modeling.

4.2 Demographics

Variable	Type	Description
age	Numeric	Age at admission (years), capped at 90 per MIMIC convention
age_group	Factor	Discretized: Adult (<65), Elderly (≥65), Very Old (≥80)
gender	Factor	Patient sex (M/F)
ethnicity	Factor	Recoded to numeric levels 0–7 (see Section 5.8)
insurance	Factor	Primary insurance type

Variables removed:

- marital_status — dropped in final cleaning
- scaled_age — dropped (multicollinear with age)

4.3 Admission & ICU Variables

Variable	Type	Description
admission_type	Factor	ELECTIVE, EMERGENCY, URGENT
first_careunit	Factor	ICU type: MICU, SICU, CCU, CSRU, TSICU
first_service	Factor	First hospital service
los_icu	Numeric	First ICU length of stay (days)
los_group	Factor	Discretized: Short (<3d), Prolonged (≥3d), Extended (≥7d)
n_icu_stays	Integer	Total ICU stays during admission
multiple_icu_stays	Binary Factor	1 if n_icu_stays > 1
is_surgical	Binary	1 if any surgical service involvement
n_service_changes	Integer	Number of service transfers
n_transfers	Integer	Number of unit transfers

Variables removed:

- admittime, dischtime, deathtime, icu_intime, icu_outtime — timestamps excluded
- scaled_los_icu — multicollinear with los_icu
- total_icu_los — redundant with los_icu
- n_unique_units — dropped in final cleaning

4.4 Survival Outcomes

Variable	Type	Description
icu_mortality	Binary	**TARGET**: 1 = died during ICU stay

Variables excluded from X (leakage/outcomes):

- event_status — hospital mortality indicator (1 = died during hospitalization)
- hospital_expire_flag — redundant with event_status
- time_to_event_hours, time_to_event_days — time-to-event (use for survival modeling only)

- end_time — death/discharge timestamp

Note: The target variable `icu_mortality` is a stricter definition than `event_status`:

- `icu_mortality = 1`: Patient died **during ICU stay** (deathtime $\leq$ icu_outtime)
- `event_status = 1`: Patient died during hospitalization (may have been after ICU discharge)

4.5 Vital Sign Summaries

Per-admission aggregates from CHARTEVENTS. **Note**: Only mean and SD statistics are retained; min/max values are dropped due to multicollinearity with mean values.

`hr_mean_val, hr_sd_val` (then removed after factorization)

Vital Sign	CareVue ID	MetaVision ID	Units
hr (heart rate)	211	220045	bpm
resp_rate	618	220210	breaths/min
spo2	646	220277	%
temp_c	676, 678→C	223762	°C
gcs	198	220739	3–15

Variables dropped due to high missingness (> 25% in demo dataset):

- sbp (systolic BP) — ∼57% missing
- dbp (diastolic BP) — ∼81% missing
- map (mean arterial pressure) — ∼58% missing

Statistics computed (intermediate, later dropped after factorization):

- `mean_val`: Average across all measurements → used to create categorical factors
- `sd_val`: Variability (instability indicator) → used for _unstable flags
- `min_val, max_val, n_obs`: **Dropped** due to multicollinearity

4.6 Laboratory Summaries

Per-admission aggregates from LABEVENTS. **Note**: Only mean values are retained and converted to categorical factors; min/max/first/last values are dropped.

Lab Test	ID	Units	Clinical Relevance
creatinine	50912	mg/dL	Renal function (SOFA)
platelets	51265	K/uL	Coagulation (SOFA)
wbc	51301	K/uL	Infection/immune
hemoglobin	51222	g/dL	Anemia, bleeding
glucose	50931	mg/dL	Metabolic status
bun	51006	mg/dL	Renal function
sodium	50983	mEq/L	Electrolyte balance
potassium	50971	mEq/L	Cardiac/renal

Labs dropped due to high missingness (> 25% in demo dataset):

- lactate (∼21% missing)

- bilirubin (∼32% missing)
- pao2, pco2 (∼29% missing)
- ph (∼26% missing)

Statistics computed (intermediate, later dropped after factorization):

- `mean_val`: Average across measurements → used to create categorical factors
- `first_val, last_val, min_val, max_val, n_labs`: **Dropped** due to multicollinearity

4.7 Comorbidity Flags (ICD-9)

Variable	Type	ICD-9 Pattern	Description
n_diagnoses	Integer	—	Total diagnosis codes
icd_sepsis	Binary Factor	995.91, 995.92, 038.x	Sepsis/septicemia
icd_heart_failure	Binary Factor	428.x	CHF
icd_aki	Binary Factor	584.x	Acute kidney injury
icd_ckd	Binary Factor	585.x	Chronic kidney disease
icd_resp_failure	Binary Factor	518.8x	Respiratory failure
icd_pneumonia	Binary Factor	480–486	Pneumonia
icd_diabetes	Binary Factor	250.x	Diabetes mellitus
icd_liver_disease	Binary Factor	571.x	Chronic liver disease
icd_malignancy	Binary Factor	140–208	Cancer
icd_stroke	Binary Factor	430–438	Cerebrovascular disease
icd_mi	Binary Factor	410.x	Myocardial infarction
icd_copd	Binary Factor	490–496	COPD
comorbidity_level	Factor	—	Low/Moderate (<3), High (≥3), Very High (≥5)

Variables removed:

- comorbidity_count — exact sum of icd_* flags (redundant)
- scaled_comorbidity — multicollinear
- scaled_n_diagnoses — multicollinear with n_diagnoses

4.8 Procedures & Interventions

Variable	Type	Description
n_procedures	Integer	Total ICD-9 procedure codes
is_ventilated	Binary Factor	Mechanical ventilation (96.7x)
received_dialysis	Binary Factor	Hemodialysis (39.95)
has_central_line	Binary Factor	Central venous catheter
received_transfusion	Binary Factor	Blood transfusion (99.0x)
n_vasopressor_doses	Integer	Total vasopressor events
vasopressor_duration_hrs	Numeric	Total hours on vasopressors
complexity_status	Factor	Standard vs. High Complexity

Variables removed:

- received_vasopressors — redundant with vasopressor_duration_hrs (> 0)
- scaled_n_procedures — multicollinear with n_procedures

4.9 DRG Severity Scores

Variable	Type	Description
drg_mortality_score	Integer	DRG mortality risk (1–4, higher = worse)
drg_severity_score	Integer	DRG severity (1–4, higher = more severe)

5 Time-Varying Matrix Z (longitudinal_df)

This dataset contains **one row per vital sign measurement occasion**. Structure: $Z_{i,t}$ = covariate vector for patient i at time t.

5.1 Key Variables

Variable	Type	Description
subject_id	Factor	Patient identifier
hadm_id	Factor	Admission identifier
icustay_id	Integer	ICU stay identifier
charttime	POSIXct	Measurement timestamp
hours_since_icu_admit	Numeric	t = hours since ICU admission (time index)
scaled_time	Numeric	Standardized time

5.2 Vital Signs (Raw Values)

Variable	Type	Description
hr	Numeric	Heart rate (bpm)
sbp	Numeric	Systolic blood pressure (mmHg)
dbp	Numeric	Diastolic blood pressure (mmHg)
map	Numeric	Mean arterial pressure (mmHg)
resp_rate	Numeric	Respiratory rate (breaths/min)
spo2	Numeric	Oxygen saturation (%)
temp_c	Numeric	Temperature (°C)
gcs	Numeric	Glasgow Coma Scale (3–15)

5.3 Scaled Vitals

`scaled_hr, scaled_sbp, scaled_dbp, scaled_map, scaled_resp_rate, scaled_spo2, scaled_temp_c`

5.4 Outcome Information (Merged from Cohort)

Variable	Type	Description
event_status	Binary	1 = died, 0 = censored
time_to_event_hours	Numeric	Patient's total survival time
end_time	POSIXct	Death or discharge timestamp
first_careunit	Factor	ICU type
age	Numeric	Age at admission
gender	Factor	Patient sex

5.5 Filtering Applied

The dataset is restricted to:
- Measurements after ICU admission (`hours_since_icu_admit` ≥ 0)
- First 7 days post-ICU admission (`hours_since_icu_admit` ≤ 168)

6 Discrete-Time Format (person_period_df)

This dataset contains **one row per patient per discrete time period**. Designed for discrete-time survival modeling with Polya-Gamma augmentation.

6.1 Why Discrete-Time Survival?

Continuous-time survival models (Cox, Weibull) can be computationally intensive in Bayesian settings. Discrete-time models offer advantages:
1. **Logistic regression formulation**: Transforms survival into sequence of binary outcomes
2. **Polya-Gamma augmentation**: Enables conjugate Gibbs sampling (no Metropolis-Hastings)
3. **Natural for time-varying covariates**: One covariate value per discrete period
4. **Computational efficiency**: Essential for large datasets like MIMIC-III

6.2 Mathematical Formulation

The discrete-time hazard:

$$h_{it} = \Pr(T_i \in (a_{t-1}, a_t] \mid T_i > a_{t-1}, X_i, Z_{i,t})$$

Logistic regression model:

$$\text{logit}(h_{it}) = \alpha_t + X_i'\beta + Z_{i,t}'\gamma + u_i$$

where:

- α_t: period-specific baseline hazard (intercept)
- X_i: static covariates (repeated across periods)
- $Z_{i,t}$: time-varying covariates (different each period)
- $u_i \sim N(0, \sigma_u^2)$: patient-level random effect (frailty)

6.3 Polya-Gamma Augmentation

The logistic likelihood is not conjugate to Gaussian priors. Polya-Gamma augmentation (Polson et al., 2013) introduces auxiliary variables $\omega_{it} \sim \text{PG}(1, \eta_{it})$ that make the conditional posterior of β Gaussian:

$$\beta \mid \omega, y \sim \text{Normal}(\cdot, \cdot)$$

This enables **fully conjugate Gibbs sampling** — dramatically improving computational efficiency and mixing for large datasets.

6.4 Key Variables

Variable	Type	Description
hadm_id	Factor	Admission identifier
subject_id	Factor	Patient identifier
period	Integer	Time period: 1, 2, ..., 7 (days)
max_period	Integer	Total periods at risk for this patient
y_it	Binary	**Discrete outcome**: 1 if event in period t
event_status	Binary	Original continuous-time indicator

6.5 Static Covariates (from X)

These are the X_i variables **repeated across all periods** for the same patient:

Variable	Type	Description
age	Numeric	Age at admission
gender	Factor	Patient sex
first_careunit	Factor	ICU type
comorbidity_count	Integer	Number of comorbidities
received_vasopressors	Binary	Any vasopressor use
is_ventilated	Binary	Mechanical ventilation

Variable	Type	Description
icd_sepsis	Binary	Sepsis diagnosis
drg_mortality_score	Integer	DRG mortality risk
los_icu	Numeric	ICU length of stay

6.6 Time-Varying Covariates (from Z)

Period-level aggregations of vital signs — these are the $Z_{i,t}$ that **change across periods**:

Variable	Type	Description
hr_mean	Numeric	Mean heart rate in period t
hr_max	Numeric	Maximum heart rate in period t
hr_n	Integer	Number of HR measurements in period t
map_mean	Numeric	Mean MAP in period t
resp_rate_mean	Numeric	Mean respiratory rate in period t
spo2_mean	Numeric	Mean SpO2 in period t
temp_c_mean	Numeric	Mean temperature in period t

6.7 Missingness Indicators

Missing vitals in ICU settings may be informative (e.g., stable patients monitored less frequently). We create indicators:

Variable	Type	Description
hr_missing	Binary	1 if no HR measurements in period (before imputation)
temp_c_missing	Binary	1 if no temperature measurements in period
map_missing	Binary	1 if no MAP measurements in period

6.8 Missing Data Imputation

For time-varying covariates in person-period format:

1. **Create missingness indicators** (before imputation) — allows model to learn if missingness is prognostic
2. **LOCF (Last Observation Carried Forward)**: Within each patient, fill missing periods with most recent available value
3. **Population median**: For patients with no prior measurements, use population median

Note: LOCF artificially shrinks variance. Consider fully Bayesian imputation (treating missing values as latent parameters) for sensitivity analysis.

7 Discretized Variables for BART Modeling

This section documents the discretization of continuous variables into **multi-level categorical factors** for BART modeling. The implementation uses `factor(case_when(...), levels = ...)` to create ordered factor variables with clinically meaningful categories.

7.1 Rationale for Multi-Level Factorization

The approach creates ordered categorical factors with 2–3 levels based on clinical thresholds:

1. **Clinical interpretability**: Categories match clinical severity grades (Normal → Mild → Severe)
2. **SOFA/SIRS alignment**: Thresholds correspond to established scoring systems
3. **Ordered structure**: Factor levels are ordered from normal to most severe
4. **BART efficiency**: Multi-level factors enable efficient tree splits at multiple thresholds
5. **Information preservation**: Distinguishes mild from severe abnormalities (vs. binary flags)

Pattern used: `factor(case_when(...), levels = c("Normal", "Mild", "Severe"))`

7.2 Variables Removed in Section 14b

The following high-cardinality and problematic variables are removed from `ready_df`:

Variable	Reason for Removal
admission_location	High-cardinality text (7+ categories with long strings)
discharge_location	High-cardinality text (15+ categories); **LEAKAGE** (known only at discharge)
language	High-cardinality character (many languages + empty strings)
religion	High-cardinality character (many religions + empty strings)
diagnosis	Free-text primary diagnosis — extremely high cardinality
dob	POSIXct datetime — use `age` instead
dod	POSIXct datetime — use `event_status` instead
expire_flag	Outcome indicator — **LEAKAGE** (redundant with event_status)
has_chartevents_data	Binary indicator — all ICU patients have chart data (no variance)

7.3 Variables Removed Due to High Missingness (Section 14)

These variables are dropped because they exceed 25% missing in the demo dataset:

Variable Pattern	Missingness	Notes
dbp_*	∼81%	Diastolic BP aggregates
map_*	∼58%	Mean arterial pressure aggregates
sbp_*	∼57%	Systolic BP aggregates
total_input_*, n_input_*	∼45%	Input volume data
n_positive_spec*, n_unique_org*	∼44%	Microbiology details
dod_hosp	∼37%	Date of death (hospital)
bilirubin_*	∼32%	Bilirubin labs
pao2_*, pco2_*	∼29%	Arterial blood gases
edregtime, edouttime	∼28%	ED timestamps
n_blood_cultures, n_positive_cult*, n_organisms	∼27%	Culture counts

Variable Pattern	Missingness	Notes
ph_*	∼26%	pH labs
lactate_*	∼21%	Lactate labs
dod_ssn	∼19%	SSN death date

Note: These thresholds are calibrated for the demo dataset. For the full MIMIC-III dataset, consider retaining variables with lower missingness.

7.4 Multi-Level Demographic Factors

Variable	Type	Levels (Ordered)	Clinical Thresholds
age_group	Factor	Adult, Elderly, Very Old	< 65, ≥ 65, ≥ 80 years
los_group	Factor	Short, Prolonged, Extended	< 3, ≥ 3, ≥ 7 days
multiple_icu_stays	Binary Factor	0, 1	n_icu_stays > 1

7.5 Multi-Level Vital Sign Factors

All thresholds based on SIRS criteria and standard clinical cutoffs. Each variable has 2–3 ordered levels.

Variable	Levels (Ordered)	Clinical Thresholds
Glasgow Coma Scale		
gcs_status	Normal, Impaired, Severe Impairment	GCS $= 15$, < 15, ≤ 8
Heart Rate		
hr_status	Normal, Bradycardia, Tachycardia	60–100, < 60, > 100 bpm
hr_unstable	0, 1 (binary factor)	HR SD > 15
Respiratory Rate		
rr_status	Normal, Tachypnea, Severe Tachypnea	≤ 20, > 20, > 30 /min
Oxygen Saturation		
spo2_status	Normal, Hypoxia, Severe Hypoxia	$\geq 95\%$, $< 95\%$, $< 90\%$
Temperature		
temp_status	Normal, Hypothermia, Fever	36–38°C, < 36°C, > 38°C
temp_unstable	0, 1 (binary factor)	Temp SD > 0.5

7.6 Multi-Level Laboratory Factors

Thresholds based on SOFA criteria and standard reference ranges.

Variable	Levels (Ordered)	Clinical Thresholds
Renal Function		
bun_status	Normal, High, Very High	≤ 20, > 20, > 40 mg/dL
creat_status	Normal, Elevated, Dysfunction	< 1.2, ≥ 1.2, ≥ 2.0 mg/dL
Glucose		
glucose_status	Normal, Hypoglycemia, Hyperglycemia	70–180, < 70, > 180 mg/dL
Hemoglobin		
hgb_status	Normal, Anemia, Severe Anemia	≥ 10, < 10, < 7 g/dL
Platelets		
plt_status	Normal, Thrombocytopenia, Severe Thrombocytopenia	≥ 150, < 150, < 50 K/uL
Electrolytes		
k_status	Normal, Hypokalemia, Hyperkalemia	3.5–5.0, < 3.5, > 5.0 mEq/L
na_status	Normal, Hyponatremia, Hypernatremia	135–145, < 135, > 145 mEq/L
White Blood Cells		
wbc_status	Normal, Leukopenia, Leukocytosis	4–12, < 4, > 12 K/uL

7.7 Multi-Level Complexity Factors

Variable	Levels (Ordered)	Clinical Thresholds
comorbidity_level	Low/Moderate, High, Very High	< 3, ≥ 3, ≥ 5 comorbidities
complexity_status	Standard, High Complexity	n_rx > 20 OR n_dx > 10 OR n_proc > 5

7.8 Binary Indicators (Converted to Factors)

The following 0/1 integer variables are converted to factors using `as.factor()`:

- **ICD Comorbidities**: icd_sepsis, icd_heart_failure, icd_aki, icd_ckd, icd_resp_failure, icd_pneumonia, icd_diabetes, icd_liver_disease, icd_malignancy, icd_stroke, icd_mi, icd_copd
- **Interventions**: is_ventilated, received_dialysis, has_central_line, received_transfusion, is_surgical
- **Medications**: on_antibiotic, on_anticoagulant, on_insulin, on_sedative
- **Cultures**: positive_blood_culture, any_positive_culture
- **Other**: callout_requested

7.9 Categorical Variables (Multi-level Factors)

These remain as multi-level factors with their original levels:

Variable	Levels
admission_type	ELECTIVE, EMERGENCY, URGENT
insurance	Medicare, Medicaid, Private, Government, Self Pay
first_careunit	MICU, SICU, CCU, CSRU, TSICU
first_service	MED, SURG, CMED, CSURG, NSURG, etc.
gender	M, F

7.10 Ethnicity Recoding

Ethnicity is recoded from text strings to numeric factor codes for consistency:

Original Value	Code	Category
WHITE	1	White
BLACK/AFRICAN AMERICAN	2	Black
ASIAN	3	Asian
AMERICAN INDIAN/ALASKA NATIVE...	4	American Indian/Alaska Native
HISPANIC OR LATINO, HISPANIC/LATINO - PUERTO RICAN	5	Hispanic/Latino
OTHER	6	Other
NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER	7	Pacific Islander
UNKNOWN/NOT SPECIFIED, OTHER/UNKNOWN, UNABLE TO OBTAIN, NA	0	Unknown

Final factor levels: `c("0", "1", "2", "3", "4", "5", "6", "7")`

7.11 Variables Removed After Factorization

After creating the categorical factors, the **original continuous values** are removed to avoid multicollinearity:

- **Original vitals**: gcs_sd_val, gcs_n_obs, hr_mean_val, hr_sd_val, hr_n_obs, resp_rate_mean_val, resp_rate_sd_val, resp_rate_n_obs, spo2_mean_val, spo2_sd_val, spo2_n_obs, temp_c_mean_val, temp_c_sd_val, temp_c_n_obs
- **Original labs**: bun_mean_val, bun_n_labs, creatinine_mean_val, creatinine_n_labs, glucose_mean_val, glucose_n_labs, hemoglobin_mean_val, hemoglobin_n_labs, platelets_mean_val, platelets_n_labs, potassium_mean_val, potassium_n_labs, sodium_mean_val, sodium_n_labs, wbc_mean_val, wbc_n_labs
- **Scaled variables**: All variables starting with `scaled_`

8 Final Cleaning and X/y Extraction (Section 17)

Section 17 of `data-prep.R` performs final cleaning to produce analysis-ready matrices for BART modeling.

8.1 Excluded Variables

The following variables are explicitly excluded from the final predictor matrix `X`:

Variable Category	Variables Excluded
Identification Variables	subject_id, hadm_id, icustay_id
Outcome Variables (Leakage Prevention)	event_status, hospital_expire_flag, time_to_event_hours, time_to_event_days
Timestamp Variables	admittime, dischtime, deathtime, icu_intime, icu_outtime, end_time
High-Cardinality/Missing Variables	discharge_location, expire_flag, dob, dod, dod_hosp, dod_ssn, diagnosis, marital_status, religion, language
Exact Multicollinearity (Scaled Duplicates)	scaled_age, scaled_los_icu, scaled_n_diagnoses, scaled_n_procedures, scaled_n_prescriptions, scaled_comorbidity
Structural Redundancy	total_icu_los (redundant with los_icu) comorbidity_count (exact sum of icd_* flags) received_vasopressors (redundant with vasopressor_duration_hrs)
EHR Frequency Artifacts	All variables ending with _n_obs and _n_labs total_output_ml, n_output_events, n_distinct_cpt, n_unique_drugs, n_unique_drug_classes

8.2 Character to Factor Conversion

All remaining character variables are converted to factors:

```
mutate_if(is.character, as.factor)
```

8.3 Missing Data Handling

Complete case analysis is applied:

```
dat <- na.omit(dat)
```

Note: This may result in sample size reduction. The script reports row counts before and after NA omission.

8.4 Final X/y and S/y_S Extraction

The target variables and predictor matrices are extracted in two steps:

Step 1: Static Matrix (X, y)

```
y <- dat$icu_mortality          # Binary outcome vector (N elements)
X <- as.data.frame(dat %>%
       select(-icu_mortality,
              -hadm_id))        # All remaining predictors (N x P)
rownames(X) <- valid_hadm_ids   # Row names = admission IDs
```

Step 2: Time-Varying Matrix (S_mat, y_S)

```
S <- person_period_df %>%
  filter(hadm_id %in% valid_hadm_ids) %>%  # Match to cleaned cohort
  select(-any_of(exclude)) %>%             # Same exclusions as X
  # ... ethnicity recoding, na.omit() ...

y_S <- S$y_it                   # Discrete-time outcome (sum T_i elements)
S_mat <- as.data.frame(S %>%
           select(-y_it,
                  -icu_mortality,
                  -hadm_id))    # Time-varying predictors
```

Key difference: `S_mat` retains the `period` variable (integer 1–7) for baseline hazard approximation, while `X` has no period column (static analysis).

8.5 Output Files

Five output files are saved to `data/rds/`:

File	Contents
ready_df.rds	Full static baseline matrix (before final exclusions)
longitudinal_df.rds	Time-varying vital sign measurements
person_period_df.rds	Discrete-time survival format (before final exclusions)
X_y.rds	**List: {X, y}** — Analysis-ready static predictors + outcome
S_y.rds	**List: {S, y}** — Analysis-ready time-varying predictors + discrete outcome

Recommended loading (via `load.R`):

```
source("load.R")
# Automatically loads: X, y, S_mat, y_S
# Use X/y for BART, logistic regression
# Use S_mat/y_S for Polya-Gamma discrete-time models
```

9 Preprocessing Rationale

9.1 Age Capping

MIMIC-III shifts dates of birth for patients >89 years (privacy protection), resulting in apparent ages >200. We cap at 90:

```
age = ifelse(age > 150, 90, age)
```

Justification: Preserves information that patient is elderly without implausible values distorting regression models.

9.2 Vital Sign Harmonization

MIMIC-III has two charting systems (CareVue and MetaVision) with different item ID schemes. We map both to common variable names.

Temperature harmonization: CareVue recorded both Celsius and Fahrenheit. We convert Fahrenheit to Celsius: $(F - 32) \times 5/9$.

9.3 Variable Scaling

Continuous predictors are initially standardized (mean=0, SD=1) in `ready_df`:

```
scaled_age = scale(age)[, 1]
```

However: In Section 17, all `scaled_*` variables are **removed** from the final `X` matrix to avoid multicollinearity with the original variables. If standardization is needed for modeling (e.g., Bayesian regression with weakly informative priors), it should be applied during model fitting.

9.4 Missing Data Defaults

- **Binary indicators**: Default to 0 (absence of code = condition not present)
- **Count variables**: Default to 0 (e.g., n_diagnoses, n_procedures)
- **DRG scores**: Default to 1 (lowest severity) — conservative choice
- **Continuous vitals/labs**: Remain NA until factorization, then patients with missing values may be excluded via `na.omit()`
- **High-missingness variables**: Dropped entirely if > 25% missing (see Section 5.2)

10 Hierarchical Structure

The data contains natural clustering that supports multi-level models:

1. **Patient level** (`subject_id`): Some patients have multiple admissions
2. **ICU level** (`first_careunit`): 5 unit types (MICU, SICU, CCU, CSRU, TSICU)
3. **Observation level**: For longitudinal_df and person_period_df, repeated measurements/periods

Example hierarchical model:

$$h_{it} = h_0(t) \exp(X_i'\beta + Z_{i,t}'\gamma + u_i + v_{k[i]})$$

where $u_i \sim N(0, \sigma_u^2)$ (patient frailty) and $v_k \sim N(0, \sigma_v^2)$ (ICU-level effect).

11 Modeling Recommendations

11.1 Suggested Workflow

1. **Exploratory**: Use `ready_df` for descriptive statistics, univariate survival curves
2. **BART modeling**: Use `X_y.rds` directly — contains pre-processed factors (e.g., *_status, *_group) for clinically interpretable tree-based analysis
3. **Latent class discovery**: BART or Dirichlet process mixtures on `X` to identify heterogeneous patient strata
4. **Discrete-time survival**: Bayesian logistic regression on `person_period_df` with Polya-Gamma augmentation
5. **Joint modeling**: Link vital trajectories (`longitudinal_df`) with survival via shared random effects

11.2 BART Implementation Notes

When fitting BART with the `BART` package in R, use the pre-processed `X_y.rds` file which has already:

1. **Excluded IDs and outcomes**: subject_id, hadm_id, icustay_id, event_status, hospital_expire_flag, time-to-event variables
2. **Excluded timestamps**: All POSIXct columns
3. **Excluded multicollinear variables**: Scaled versions, redundant aggregates
4. **Converted to factors**: All categorical and discretized variables
5. **Applied complete-case analysis**: NA rows removed
6. **Recoded ethnicity**: Consolidated to 8 levels (0–7)

Key variables in final X matrix:

- **Multi-level factors**: age_group, los_group, gcs_status, hr_status, rr_status, spo2_status, temp_status, bun_status, creat_status, glucose_status, hgb_status, plt_status, k_status, na_status, wbc_status, comorbidity_level, complexity_status
- **Binary factors**: hr_unstable, temp_unstable, multiple_icu_stays, all icd_* flags, intervention indicators
- **Categorical factors**: admission_type, insurance, first_careunit, first_service, gender, ethnicity
- **Numeric**: age, los_icu, n_diagnoses, n_procedures, n_prescriptions, drg_mortality_score, drg_severity_score, n_vasopressor_doses, vasopressor_duration_hrs, n_service_changes, n_transfers, n_callouts, gcs_mean_val

Example R code for BART:

```
# Load pre-processed data
data <- readRDS("data/rds/X_y.rds")
X <- data$X
y <- data$y

# Fit logistic BART for binary outcome (icu_mortality)
library(BART)
bart_model <- lbart(x.train = X, y.train = y)
```

11.3 Prior Recommendations

Note: The final `X` matrix primarily contains factor variables (both multi-level and binary). For any remaining numeric predictors (age, los_icu, count variables), consider standardizing during model fitting.

For numeric predictors (when standardized):

$$\beta_j \sim \text{Normal}(0, 2.5)$$
$$\sigma_u \sim \text{Half-Cauchy}(0, 2.5)$$

For period-specific intercepts in discrete-time models (baseline hazard):

$$\alpha_t \sim \text{Normal}(0, 2) \quad \text{or random walk prior}$$

For BART models, use default priors — BART's regularization priors are well-calibrated for most applications.

12 Data Quality Notes

1. **NOTEEVENTS**: Empty in demo version (no clinical notes)
2. **Age >89**: Capped at 90 due to MIMIC de-identification
3. **ICD-9 codes**: Assigned retrospectively for billing; may not reflect status at ICU admission
4. **Dual charting**: CareVue and MetaVision merged via item ID mapping
5. **Lab missingness**: May be informative (stable patients get fewer labs)
6. **LOCF limitations**: Artificially reduces variance; use missingness indicators as covariates
7. **Demo vs. Full**: Demo has ∼100 patients; full has ∼46,000